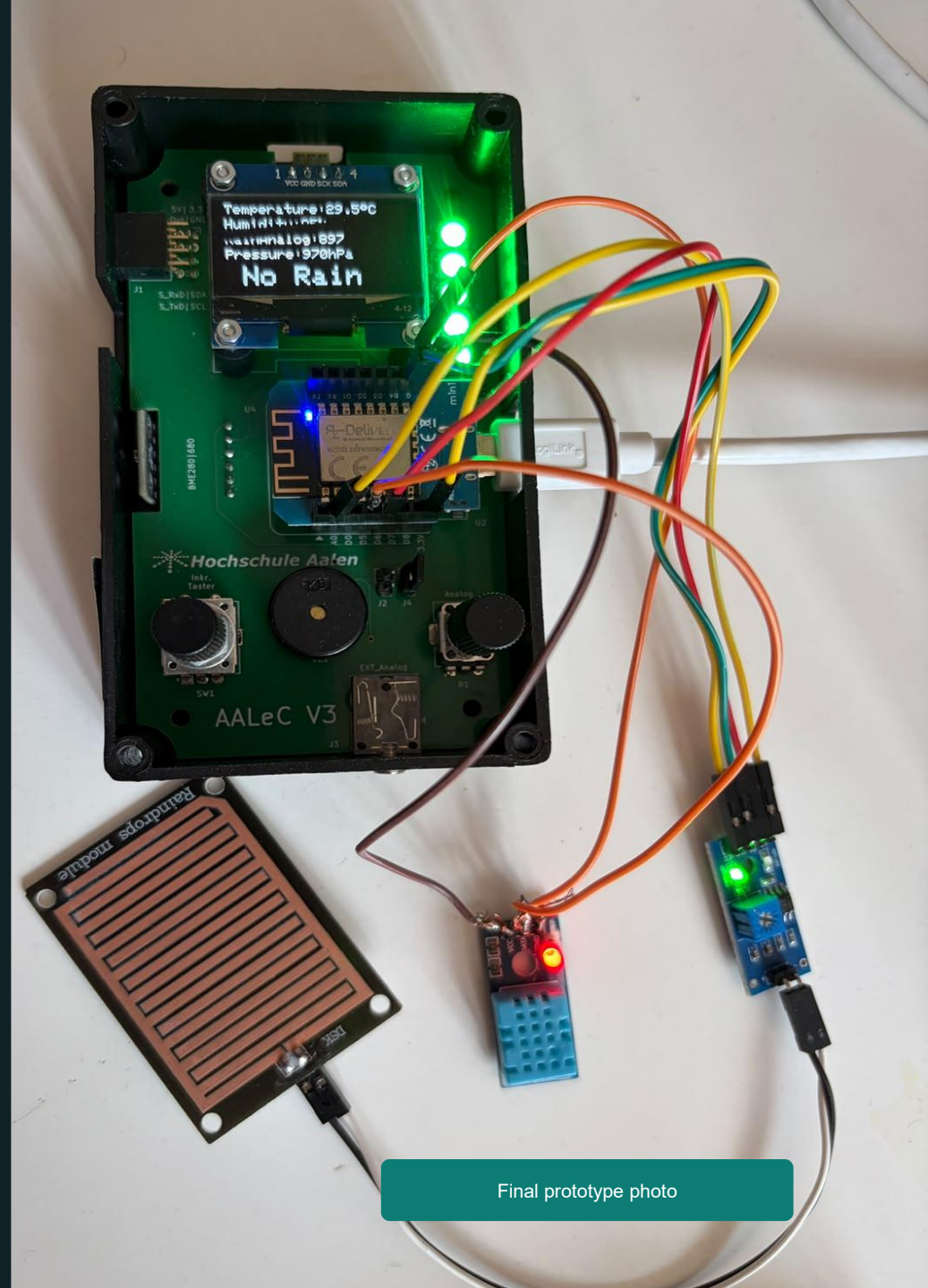


PROGRAMMING INTERNET OF THINGS

IoT-Based Room Climate & Rain Monitoring System

ESP8266 + DHT11 + Rain Sensor + MQTT +
OLED + Buzzer + LEDs

Final project presentation



Final prototype photo

Use case and project idea

Why this project?

💧 Humidity problem

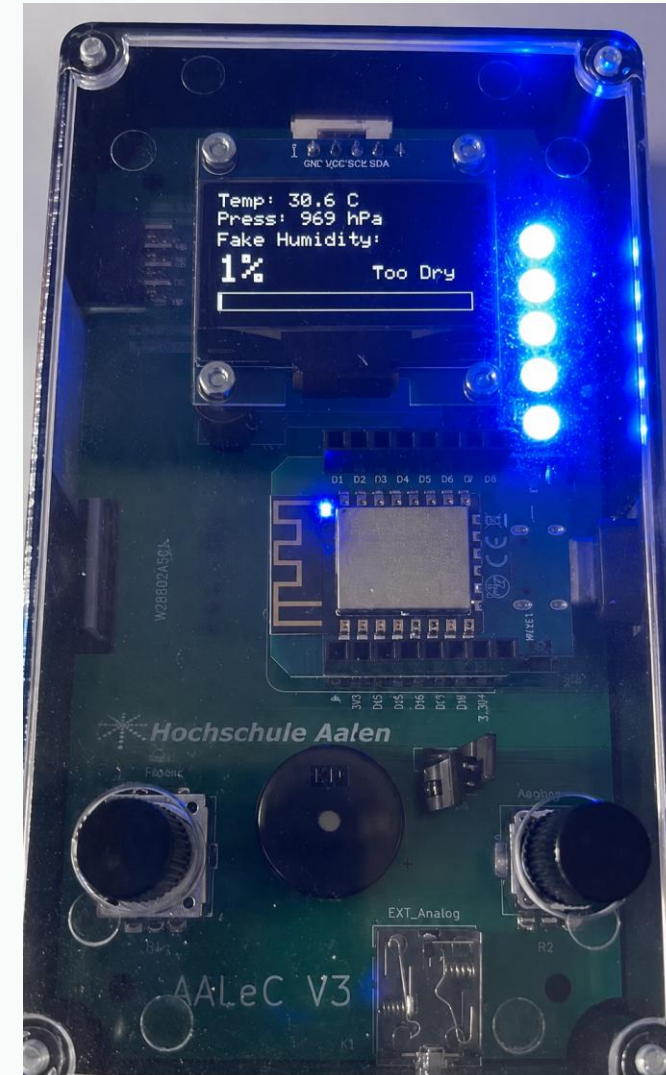
Indoor humidity can become a problem in dorms, basements, and poorly ventilated rooms.

⚠️ Risk if unnoticed

Moisture may lead to mold risk, uncomfortable room conditions, or hidden water leakage.

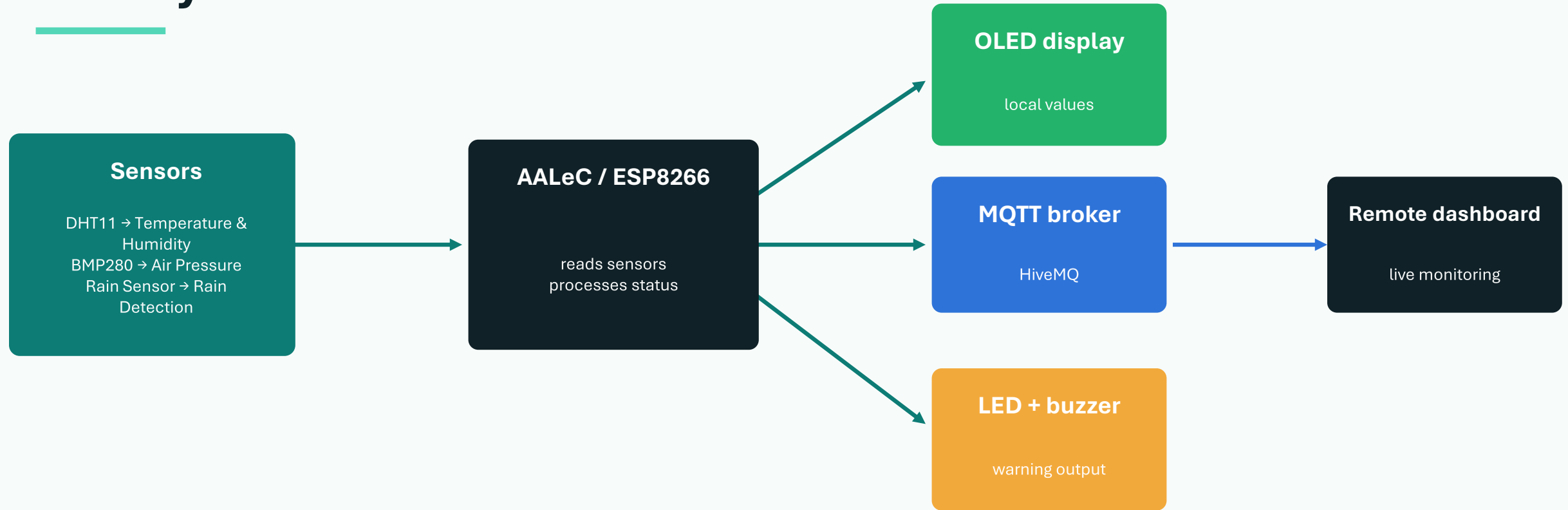
✅ Our solution

The project monitors room climate (temperature, humidity, pressure) and detects rain presence in real time using sensors, OLED feedback, LEDs, buzzer, and MQTT.



Initial prototype with fake humidity

Final system overview

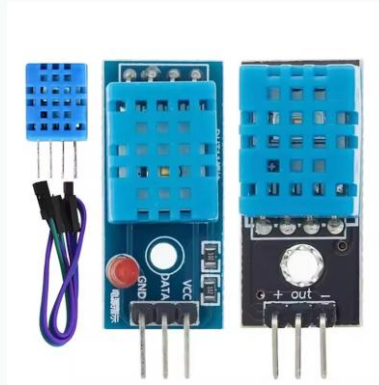


Local feedback and remote monitoring run in parallel: the device remains useful even without the dashboard.

Hardware components



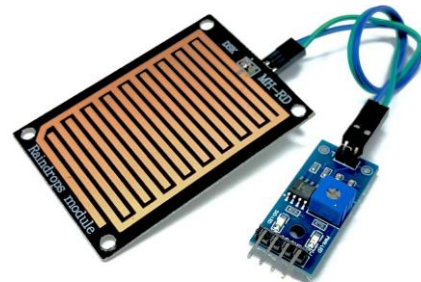
Main AALeC V3 board



DHT11



BMP280 on the board



Raindrops Module

- AALeC V3 board with ESP8266 controller
- OLED display for local values and status
- DHT11 module for temperature and humidity
- BMP280 pressure sensor for atmospheric pressure
- Raindrops Module for dry / raining status
- RGB LEDs and buzzer for visual/acoustic warning

How the sensors work

DHT11

sends temperature and humidity's digital data

DHT11 measures temperature and relative humidity. It contains a humidity element and a thermistor, then outputs digital values readable by the controller.

BMP280

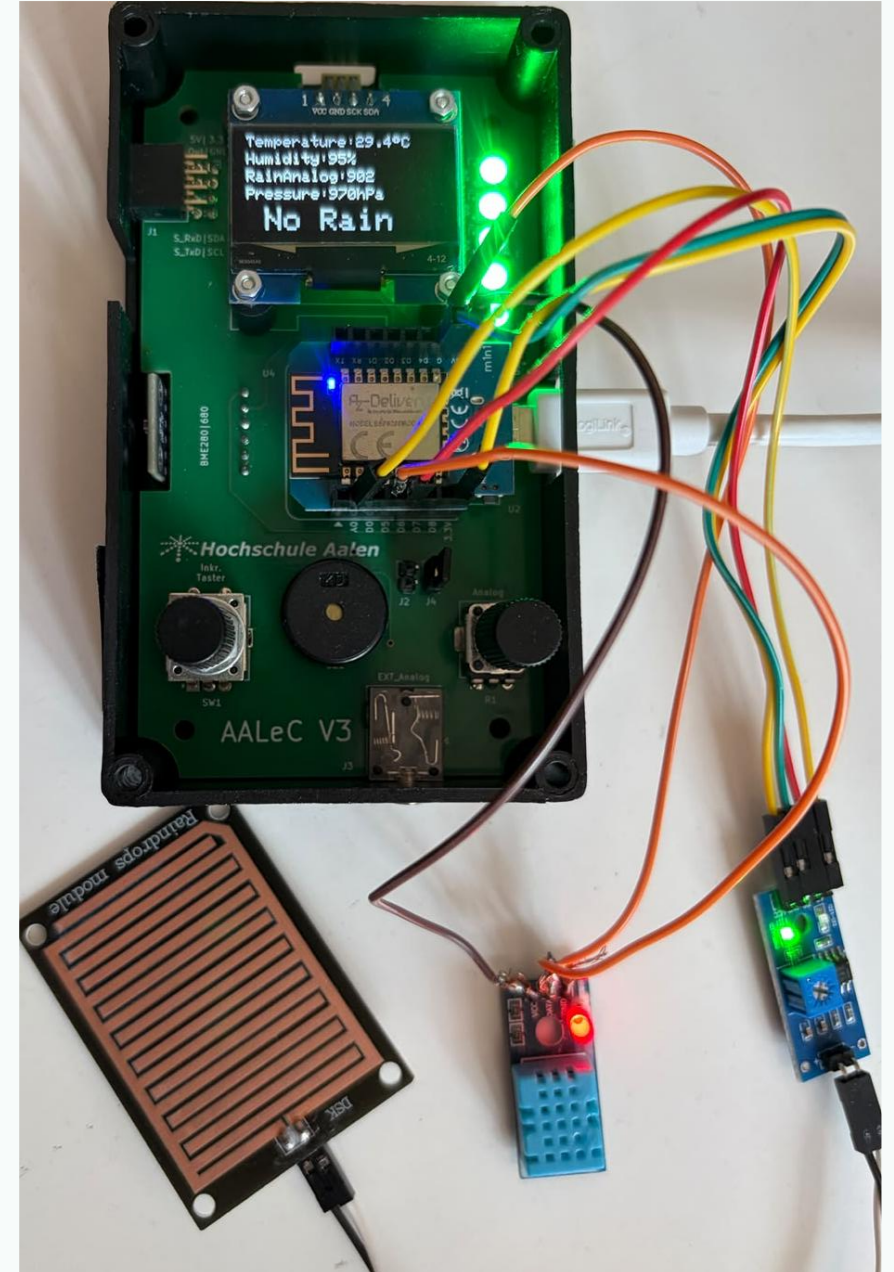
pressure sensing element
pressure changes electrical response

The BMP280 pressure sensor converts atmospheric pressure changes into a digital pressure value. Only pressure is used from this sensor in the final system.

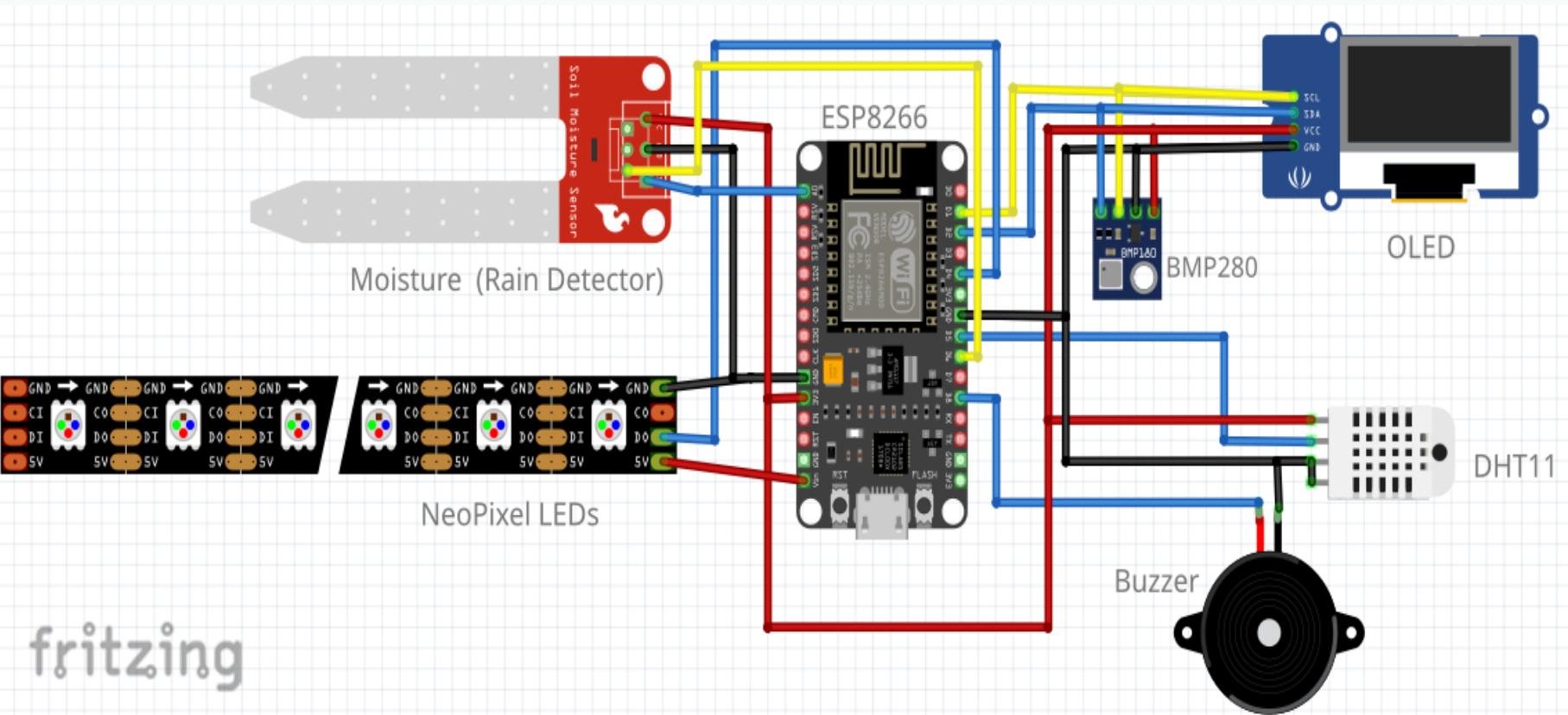
RAINDROPS SENSOR

raindrops module
analog + digital output

The raindrop sensor detects droplets through resistance changes. This module gives an analog moisture value and a digital Dry/Rain decision.

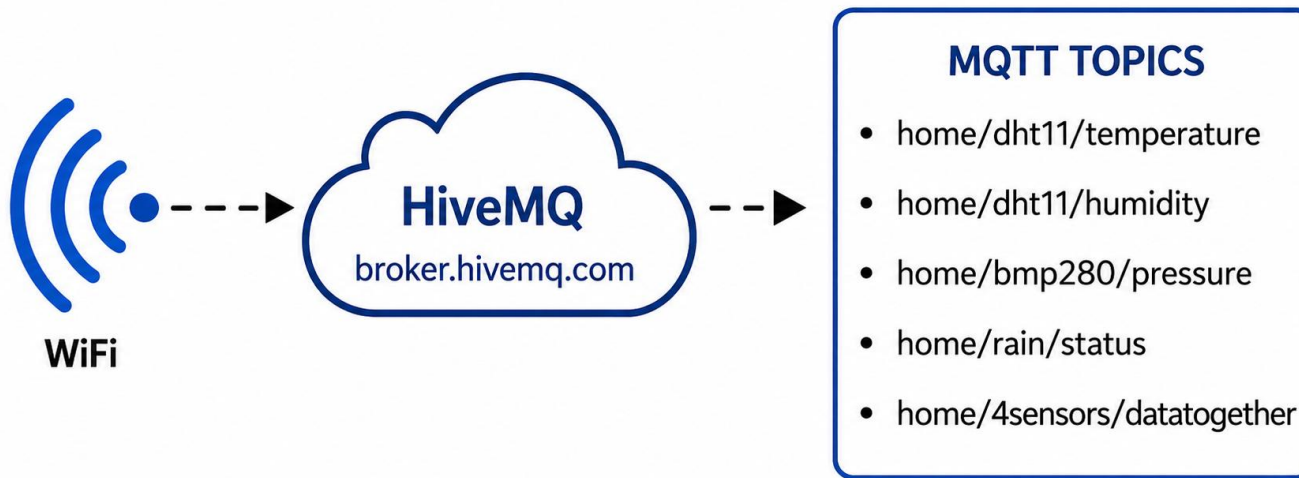


Wiring and interfaces



Device	Pin
DHT11	D5
Rain Digital	D6
Rain Analog	A0
BMP280	D1/SCL; D2/SDA
Buzzer	D8 (GPIO15)
NeoPixel LEDs	D4 (GPIO2)
OLED	I ² C

Network connection and Libraries:



- ESP8266 connects to Wi-Fi and an MQTT broker.
- Sensor values are published to project topics.
- HiveMQ WebSocket Client or another MQTT client can subscribe and show live values.
- This makes the project testable from another device on the network.

Libraries Used to communicate:

- ESP8266WiFi
- PubSubClient
- DHT
- Adafruit BMP280
- Adafruit GFX
- Adafruit SH110X
- Adafruit NeoPixel

Setup and configuration

1 Connect hardware

DHT11, rain sensor, OLED, pressure sensor, LED/buzzer on the AALeC board.

2 Configure network

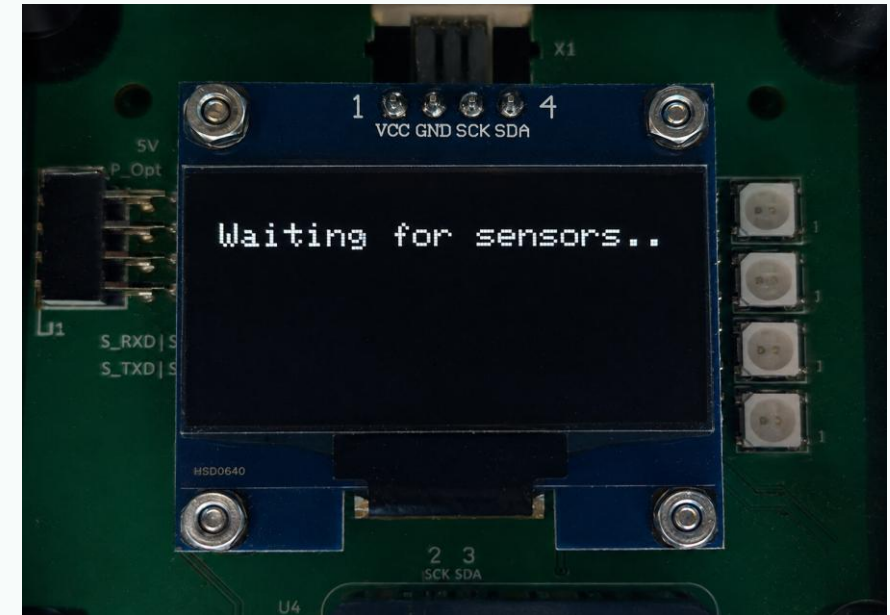
Set Wi-Fi SSID/password and MQTT broker address in the code.

3 Upload and test

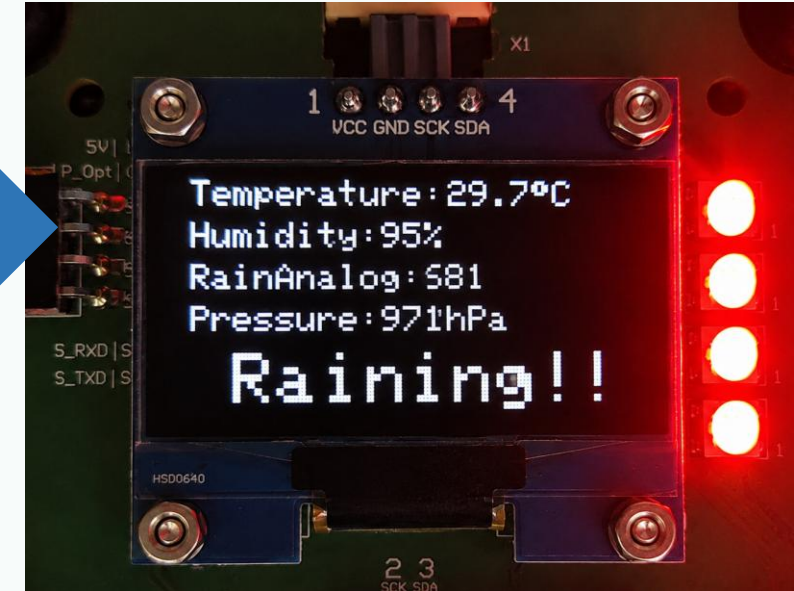
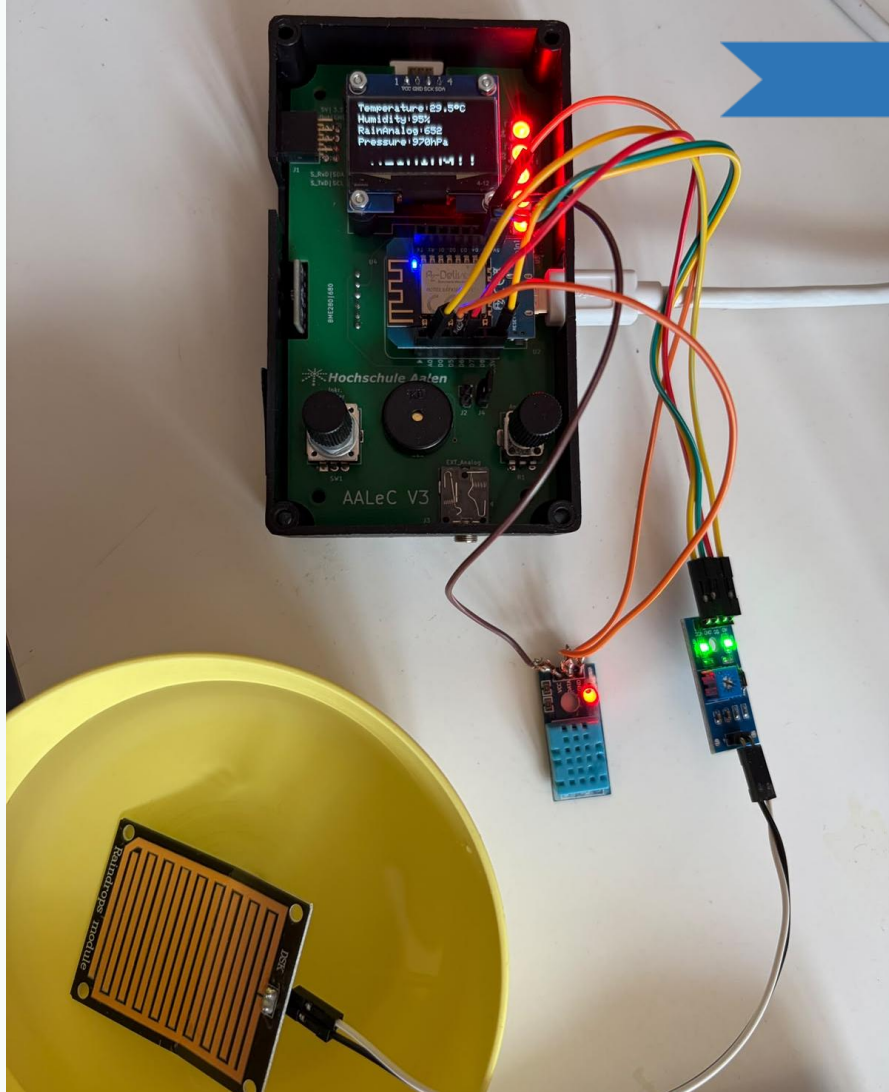
Upload to ESP8266, open serial monitor, then subscribe to MQTT topics.

4 Demonstrate functionality

Show OLED output, rain sensor response, LED/buzzer behavior and MQTT dashboard.



Local display and warning behavior



The OLED and LEDs show the current state directly on the device, while Serial Monitor / MQTT are used for debugging and proof of data flow.

Dry:

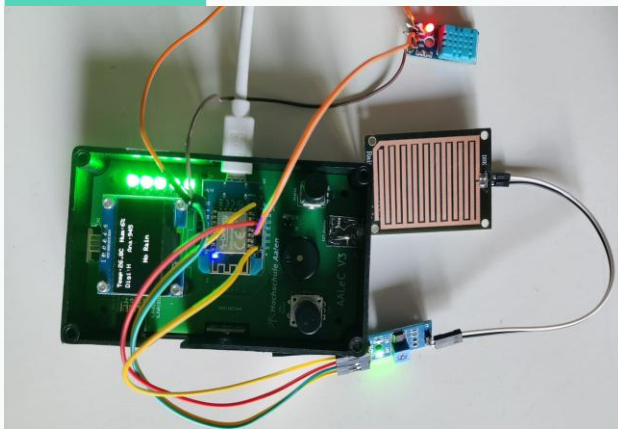
- Green LEDs ON
- OLED shows "No Rain"

Rain Detected:

- Red LEDs blinking
- Buzzer ON
- OLED shows "Raining!!" Analog rain value allows more detailed moisture intensity observation.

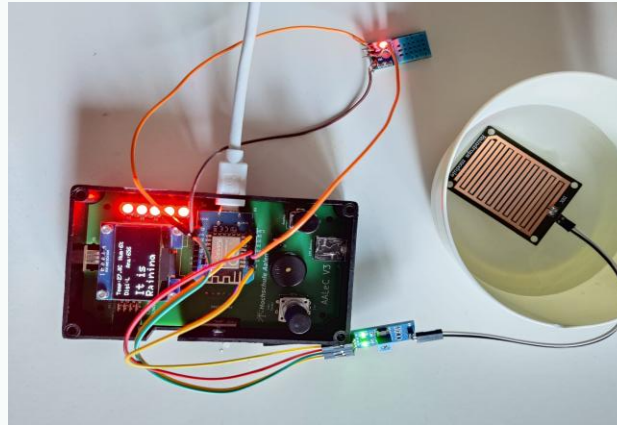
Analog rain value allows more detailed observation.

Final result



Dry / normal state

- OLED shows values and No Rain.
- Green LEDs indicate normal condition.
- Analog rain value remains high.



Raining detected

- Water on the rain module lowers the analog value.
- Digital state changes to rain.
- OLED and LEDs switch to warning state.



```

Temperature: 29.9 °C | Humidity: 95% | Air Pressure: 969.9 hPa | Rain Analog: 902 | Digital State: Dry | Rain Status: No Rain | Alarm OFF
Temperature: 29.9 °C | Humidity: 95% | Air Pressure: 969.9 hPa | Rain Analog: 902 | Digital State: Dry | Rain Status: No Rain | Alarm OFF
Temperature: 29.9 °C | Humidity: 95% | Air Pressure: 969.9 hPa | Rain Analog: 902 | Digital State: Dry | Rain Status: No Rain | Alarm OFF
Temperature: 29.9 °C | Humidity: 95% | Air Pressure: 969.9 hPa | Rain Analog: 902 | Digital State: Dry | Rain Status: No Rain | Alarm OFF
Temperature: 29.8 °C | Humidity: 95% | Air Pressure: 969.9 hPa | Rain Analog: 902 | Digital State: Dry | Rain Status: No Rain | Alarm OFF
Temperature: 29.9 °C | Humidity: 95% | Air Pressure: 969.9 hPa | Rain Analog: 897 | Digital State: Dry | Rain Status: No Rain | Alarm OFF
  
```

Evidence and next steps

- Serial Monitor verifies the transition from Dry to Rain.
- MQTT can publish live values for remote monitoring.

Connection

connected

Publish

⌵

Messages

⌵

Subscriptions

Add New Topic Subscription

Qos: 2

home/4sensors/data...

2026-06-20 21:13:45

Topic: home/4sensors/datatogether

Qos: 0

Temperature=29.2C | Humidity=95% || Pressure=971.1hPa | No Rain

2026-06-20 21:13:43

Topic: home/4sensors/datatogether

Qos: 0

Temperature=29.0C | Humidity=95% || Pressure=971.1hPa | No Rain

2026-06-20 21:11:34

Topic: home/4sensors/datatogether

Qos: 0

Temperature=29.2C | Humidity=95% || Pressure=971.1hPa | Rain

2026-06-20 21:11:34

Topic: home/4sensors/datatogether

Qos: 0

Temperature=29.2C | Humidity=95% || Pressure=971.1hPa | Rain

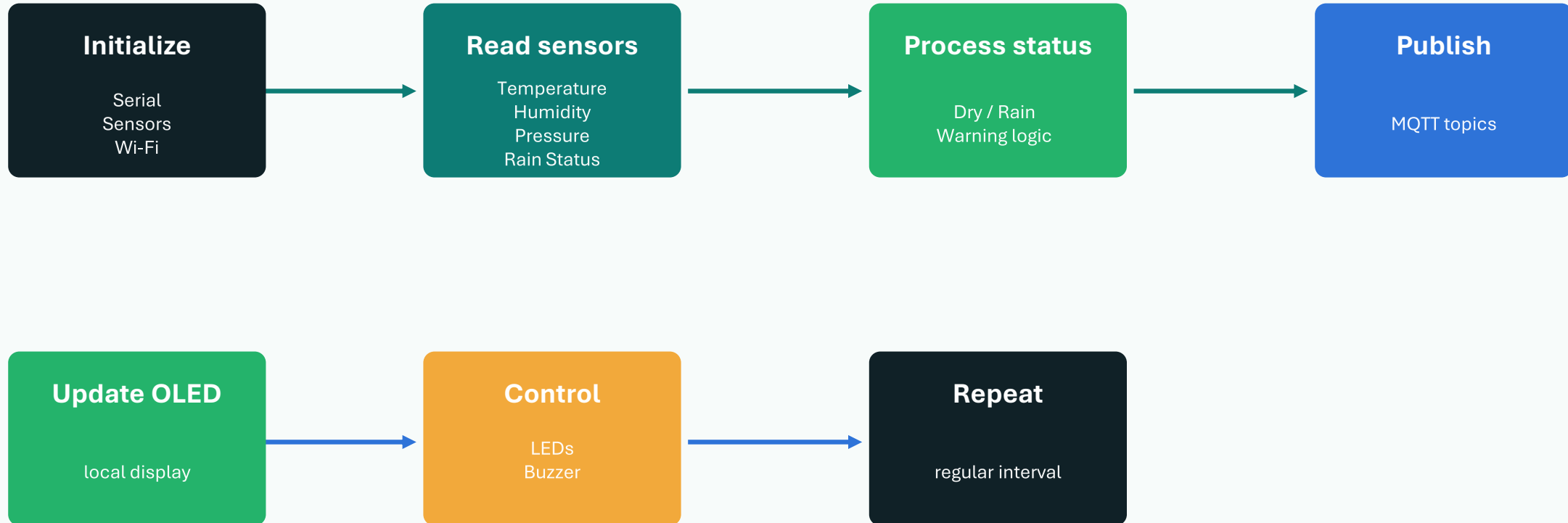
2026-06-20 21:11:30

Topic: home/4sensors/datatogether

Qos: 0

Temperature=29.3C | Humidity=95% || Pressure=971.1hPa | Rain

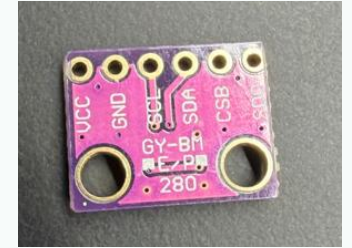
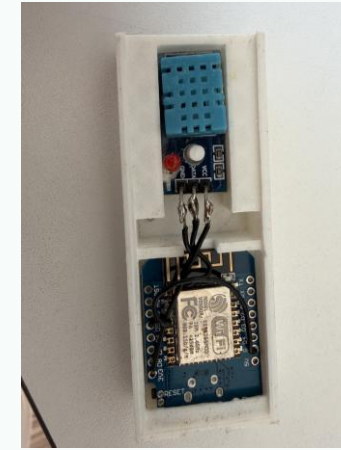
Software stack and code structure



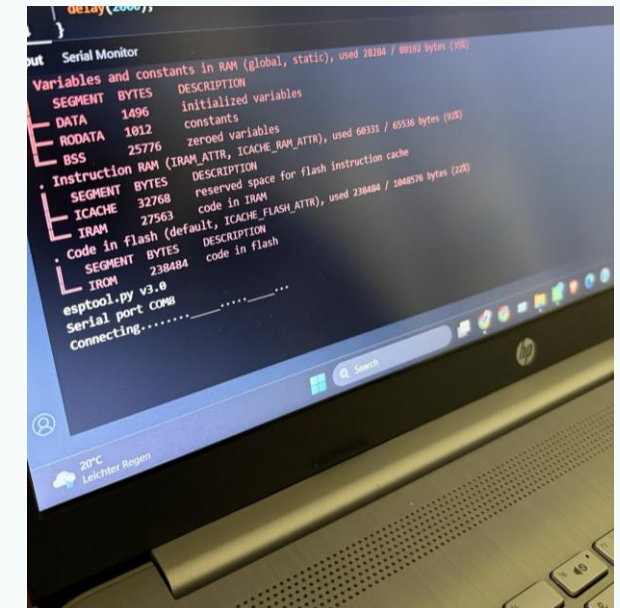
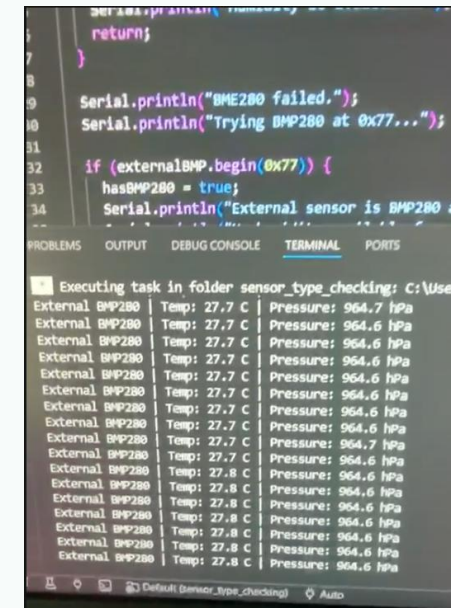
The Arduino/ESP8266 code continuously collects readings, checks the rain state, updates the OLED and publishes values to MQTT at regular intervals.

Problems and workarounds

Implementation issues were part of the project work, so the final documentation should explain them clearly.

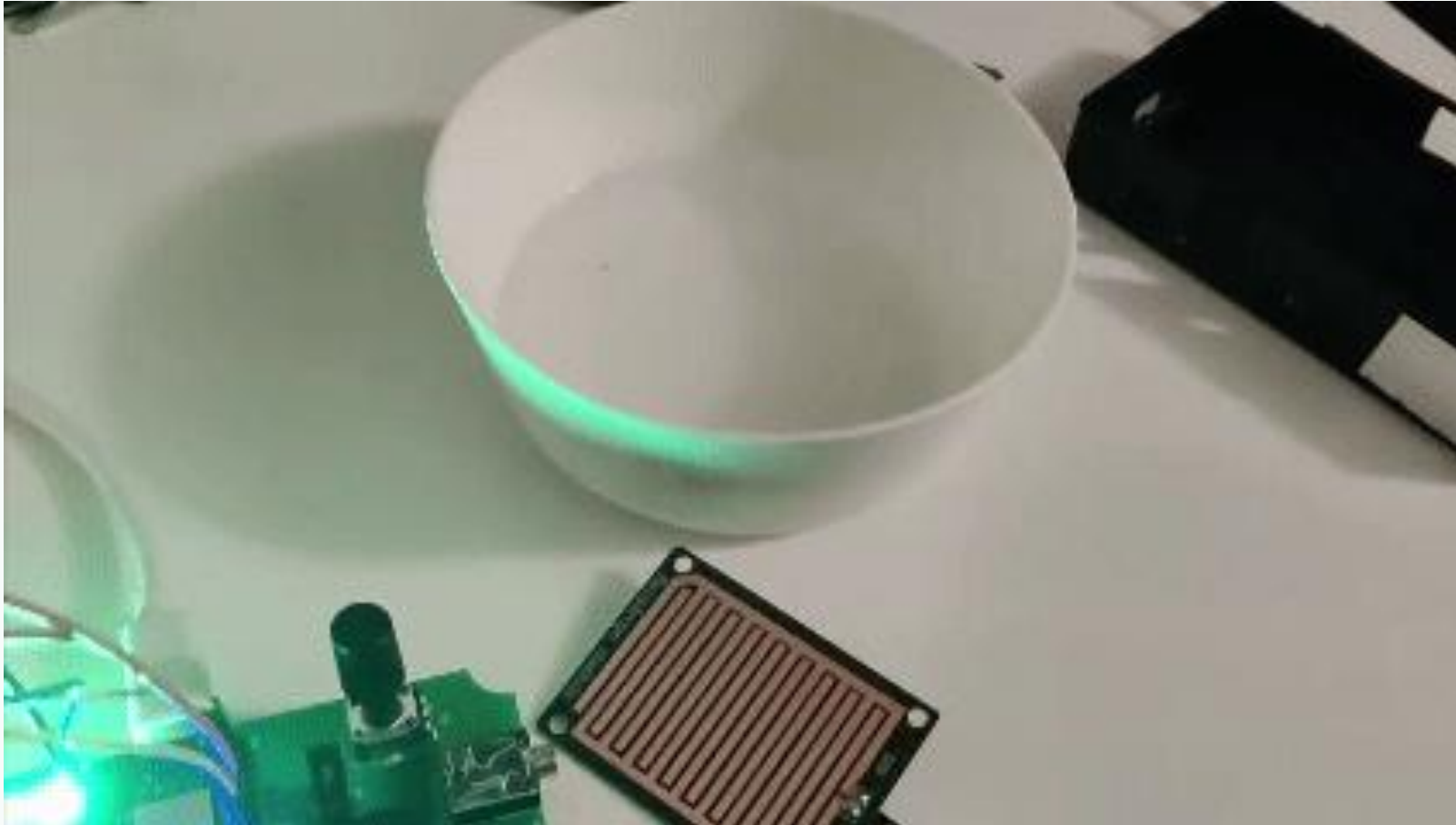


- OLED initially showed noisy pixels; correct SH1106 library fixed the display.
- External BMP/BME module was detected as BMP280, so it did not provide humidity.
- The DHT11 sensor was not functioning when it was linked to the Wi-Fi module (it did not appear in the COM port during testing), so we had to disassemble it to connect it directly to the ESP8266 module.
- AALeC pin conflicts caused failures; DHT11 data was transferred to D5 after individually testing each pin.
- Upload/COM port issue was resolved by individually testing the ESP8266 and identifying that the resistor in AaLeC v3 is the cause of the problem.



Here are a few images depicting some issues we encountered during the prototype phase.

Demonstration and Conclusion



Conclusion

- Successfully monitored temperature, humidity, pressure, and rain status
- Displayed data locally on OLED
- Generated visual and audible alerts
- Published live data to HiveMQ using MQTT
- Demonstrated both local and remote monitoring

Future improvements

- More accurate humidity sensor
- Weather API integration
- Cleaner enclosure/wiring
- Historical dashboard storage